Case Study 2: Student Marks and Grade Calculator

1. Problem Statement

Read a student's marks in three subjects, calculate the total and average, and assign a grade based on the average marks.

2. Objectives

1. Read student name and three subject marks.
2. Calculate total and average marks.
3. Use conditional statements to assign a grade.
4. Print a marks statement.

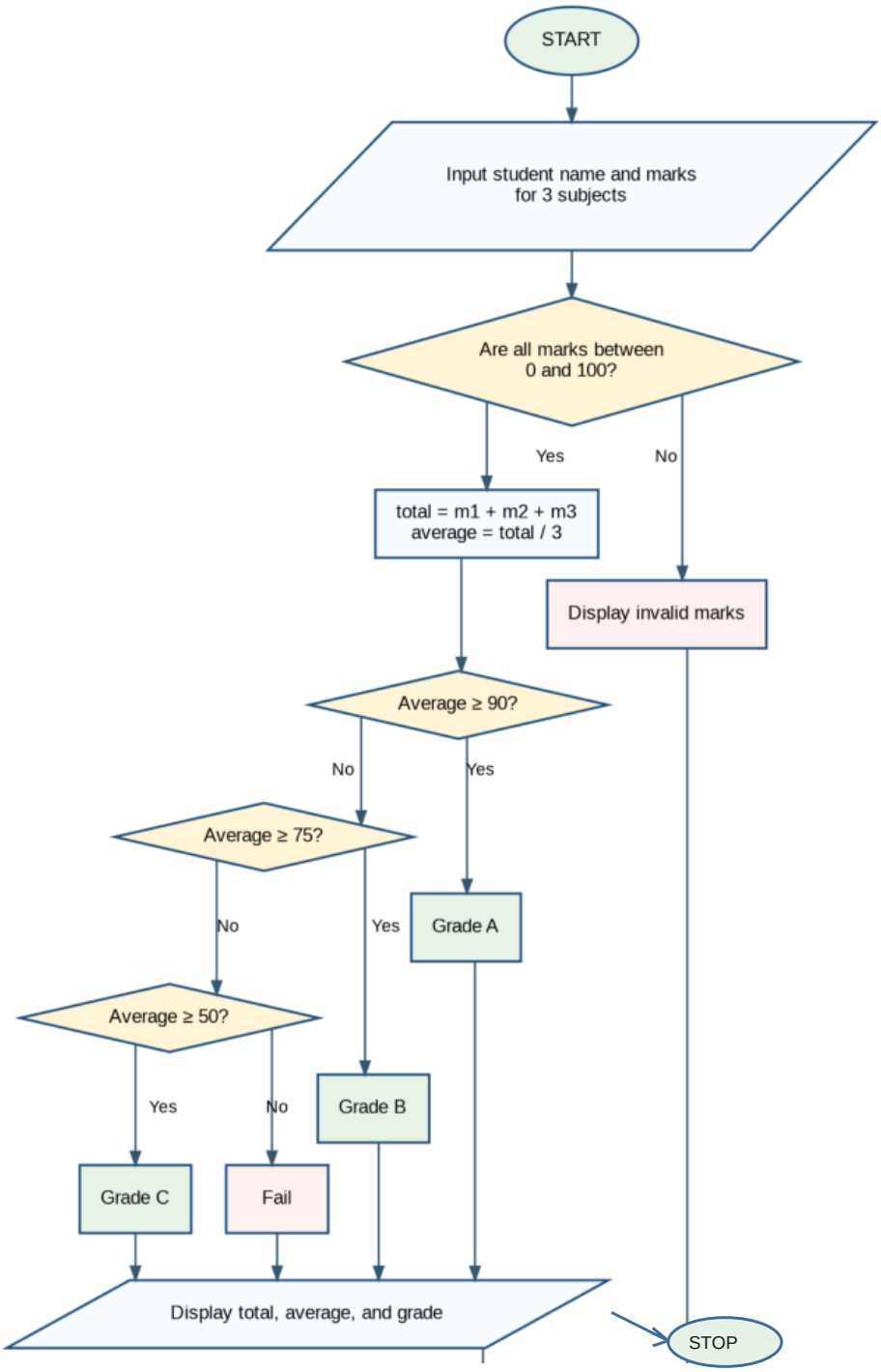
3. Assumptions

1. Each subject is out of 100 marks.
2. Marks must be between 0 and 100.
3. Grade A: 90 and above; B: 75–89; C: 50–74; Fail: below 50.

4. Algorithm

1. Start.
2. Read student name and marks in three subjects.
3. Validate that each mark is between 0 and 100.
4. Calculate total and average.
5. Use if-elif-else to assign a grade.
6. Display the marks statement.
7. Stop.

5. Flowchart



6. Pseudocode

```
INPUT name, m1, m2, m3
 total = m1 + m2 + m3
 average = total / 3
IF average >= 90 THEN grade = "A"
ELSE IF average >= 75 THEN grade = "B"
ELSE IF average >= 50 THEN grade = "C"
ELSE grade = "Fail"
DISPLAY total, average, grade
```

7. Complete Python Source Code

```
name = input("Enter student name: ")
m1 = float(input("Enter marks in Subject 1: "))
m2 = float(input("Enter marks in Subject 2: "))
m3 = float(input("Enter marks in Subject 3: "))
if any(mark < 0 or mark > 100 for mark in [m1, m2, m3]):
    print("Invalid marks")
else:
```

```
total = m1 + m2 + m3
average = total / 3
if average >= 90:
    grade = "A"
elif average >= 75:
    grade = "B"
elif average >= 50:
    grade = "C"
else:
    grade = "Fail"
print("\n----- MARKS STATEMENT -----")
print("Student:", name)
print("Total:", total, "/ 300")
print(f"Average: {average:.2f}")
print("Grade:", grade)
```

8. Test Case and Expected Output

Name = Anu; Marks = 85, 78, 82.

Total = 245; Average = 81.67.

Expected grade = B.

9. Conclusion

The program calculates total and average marks and assigns a grade using if-elif-else.